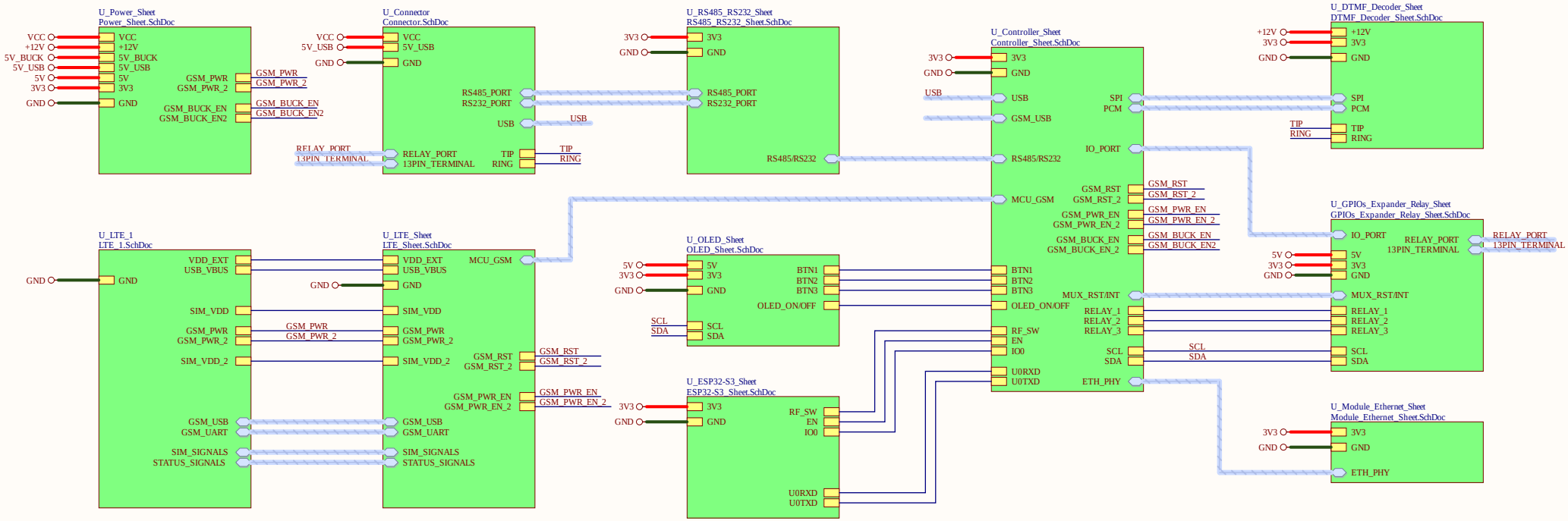


Option 1: (Single Path - Ethernet capability only)
---> DNP U13, U16, U17, J21, J23.

Option 2: (Single Path - LTE capability only)
---> DNP J20, U12.

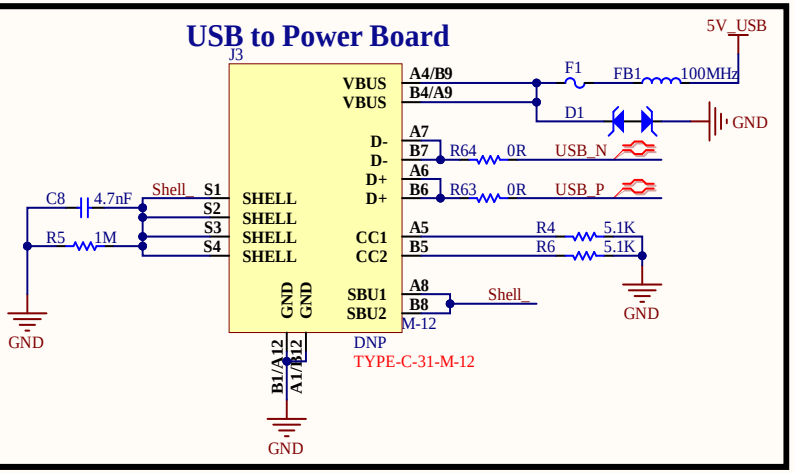
Option 3: (Dual Path - LTE and Ethernet both)
---> DNP U17.

Option 4: (Dual Path - 2 x LTE)
---> DNP J20, U12.

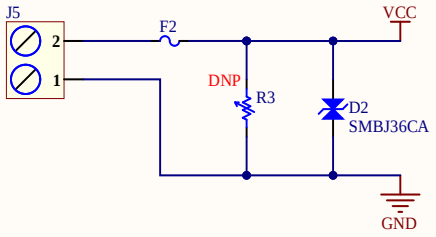
eSpark Consulting Group

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Date: 14/03/2024	Page	Num: 1
Drawn By: Huzaifa Ali	Page	Total: 12

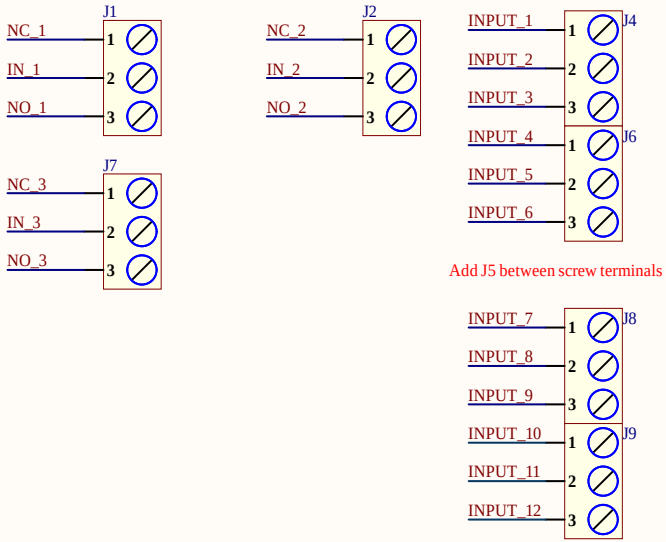
USB to Power Board



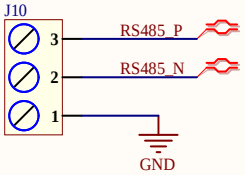
9-30V DC INPUT



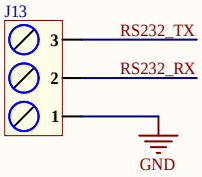
SCREW TERMINALS



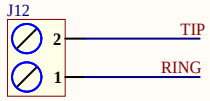
PORT for RS485

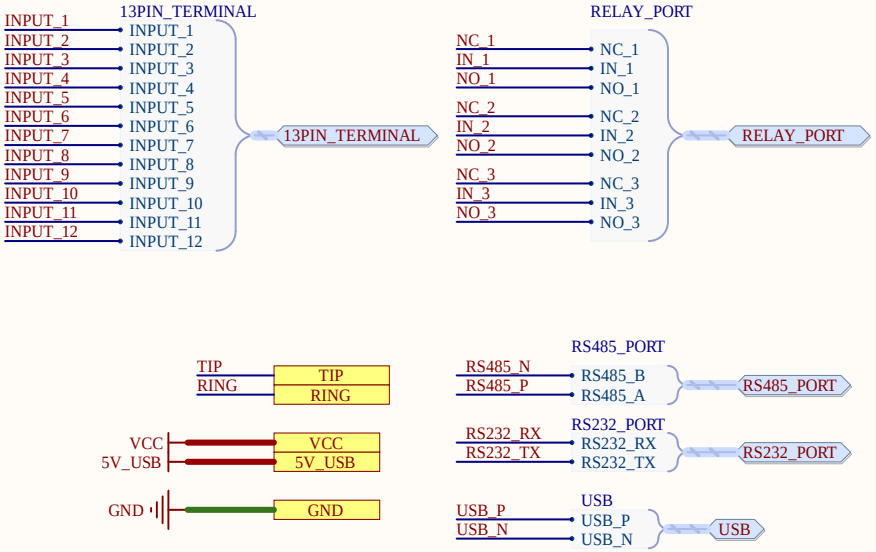


PORT for RS232



PORT for DTMF

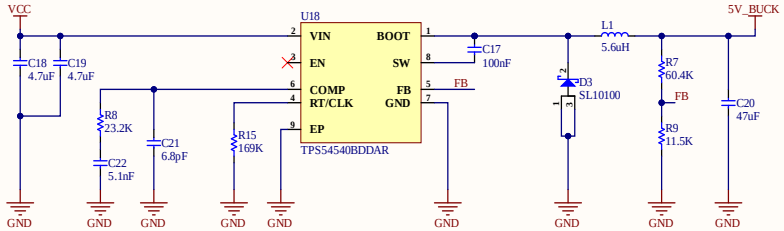




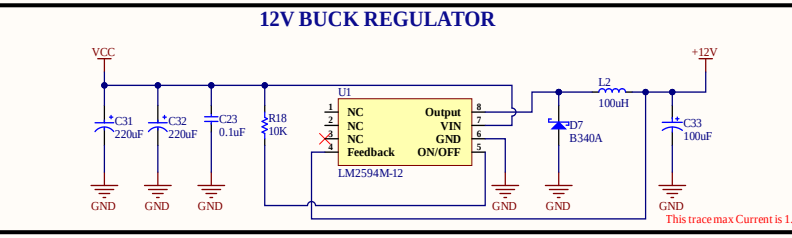
eSpark Consulting Group

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Date: 14/03/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 12

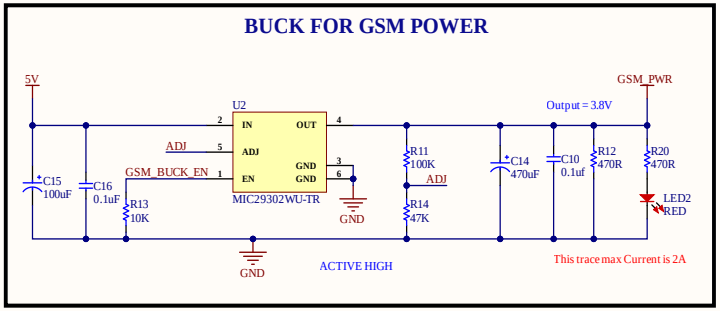
5V BUCK REGULATOR



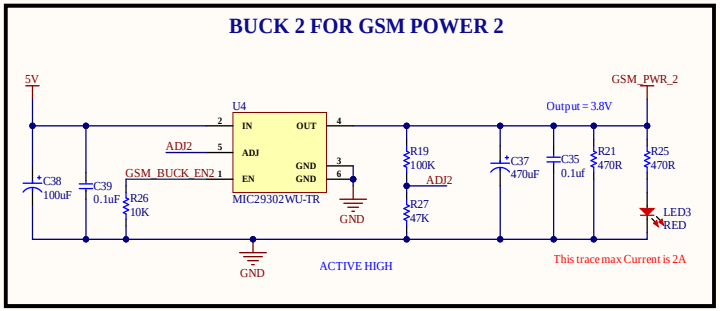
12V BUCK REGULATOR



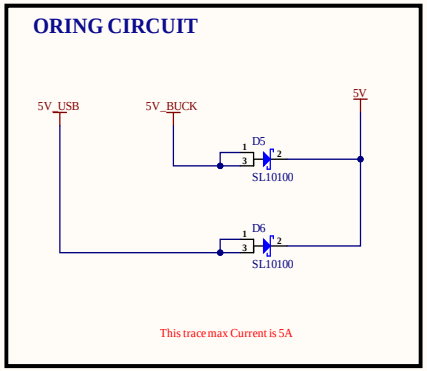
BUCK FOR GSM POWER



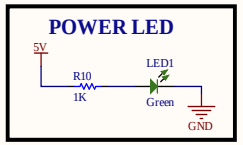
BUCK 2 FOR GSM POWER 2



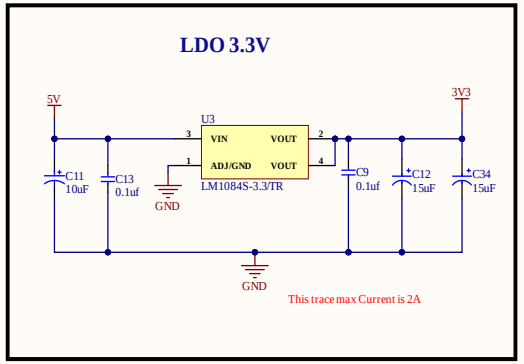
ORING CIRCUIT



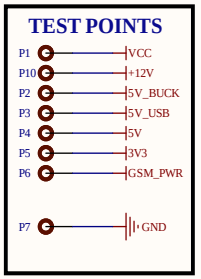
POWER LED

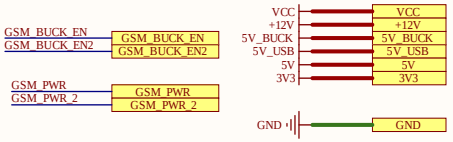


LDO 3.3V



TEST POINTS

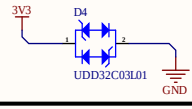




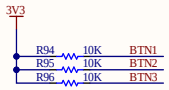
eSpark Consulting Group

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Date: 15/03/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 12

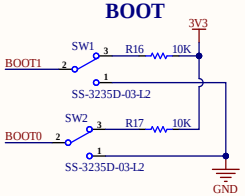
ESD PROTECTION



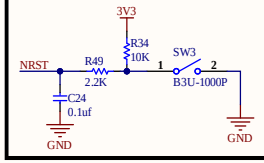
PULL-UP



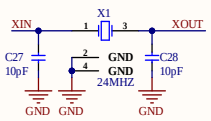
BOOT



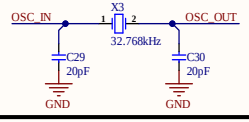
RESET

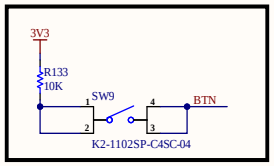


24MHz CRYSTAL

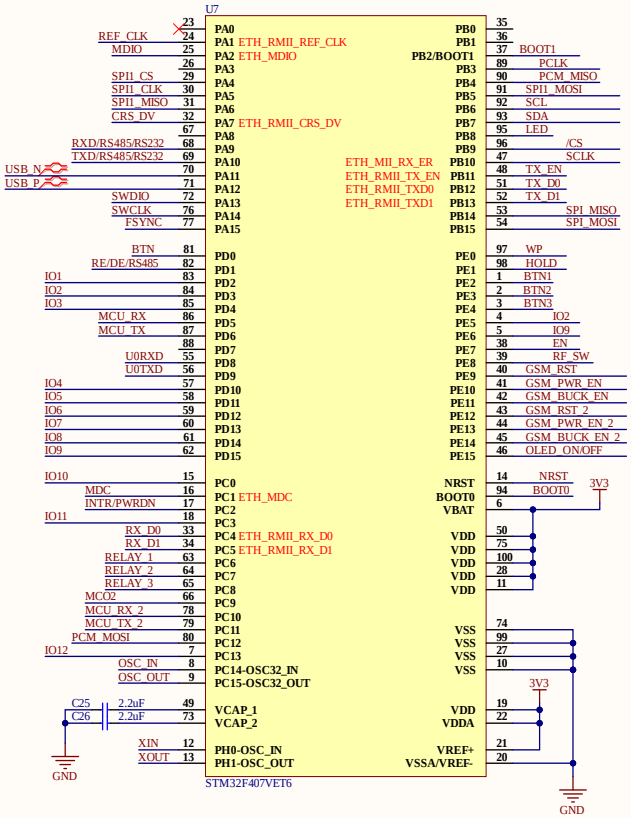


32.768KHZ CRYSTAL

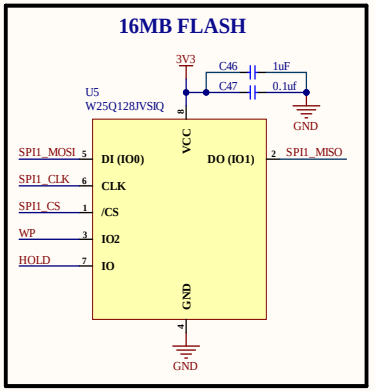


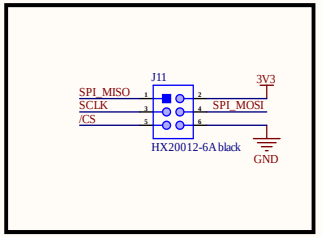


MCU

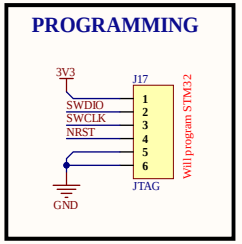


16MB FLASH

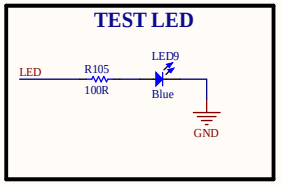


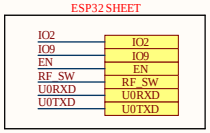


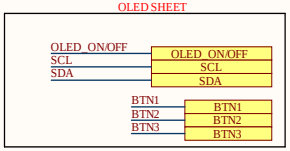
PROGRAMMING

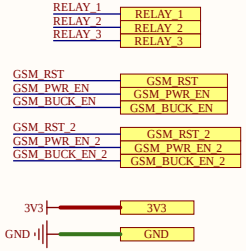


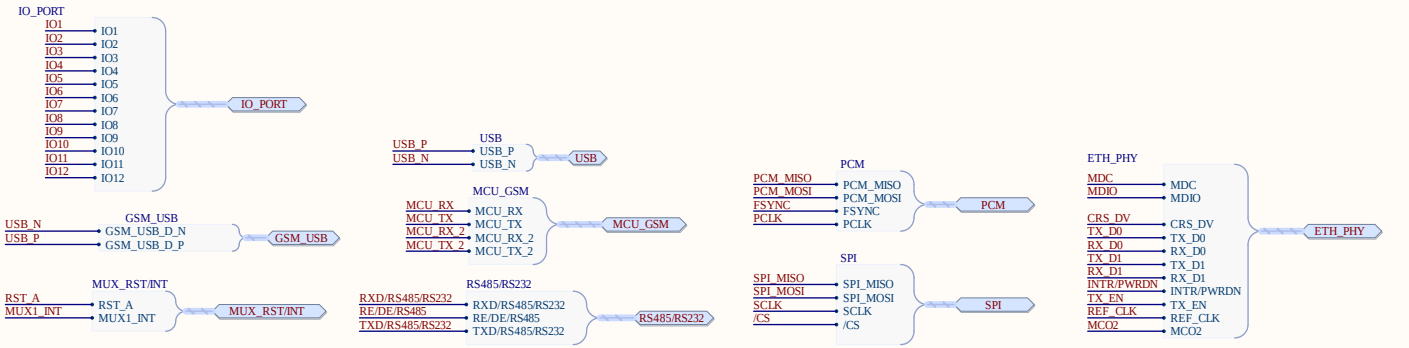
TEST LED





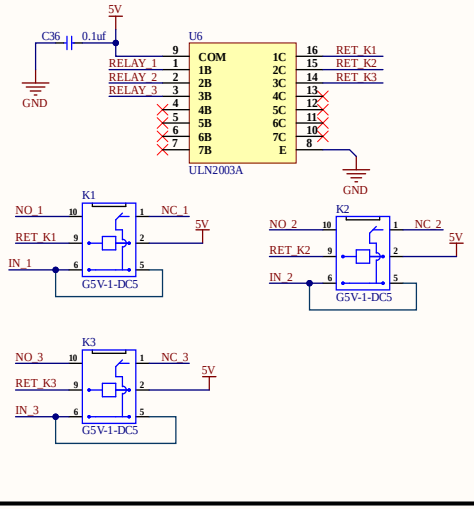




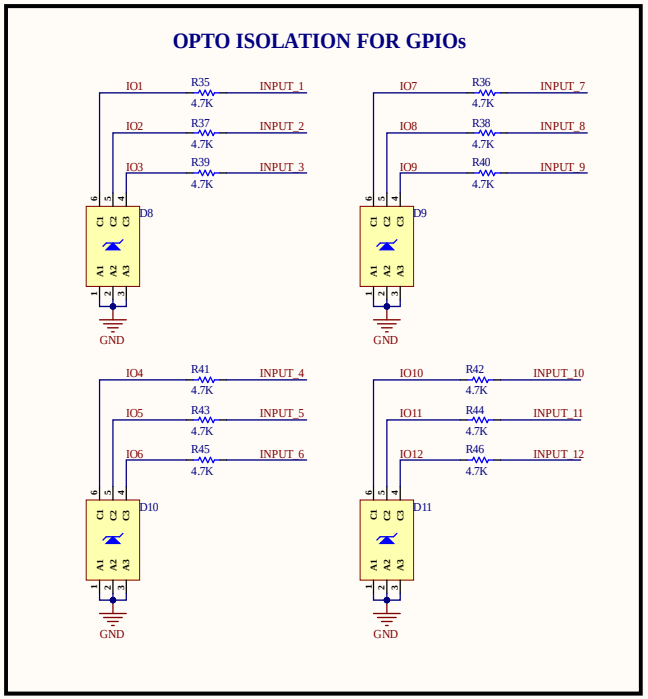


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Drawn By: Huzaifa Ali	Page Total: 12

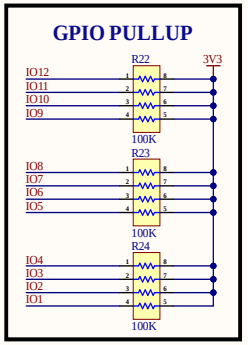
RELAY WITH DRIVER

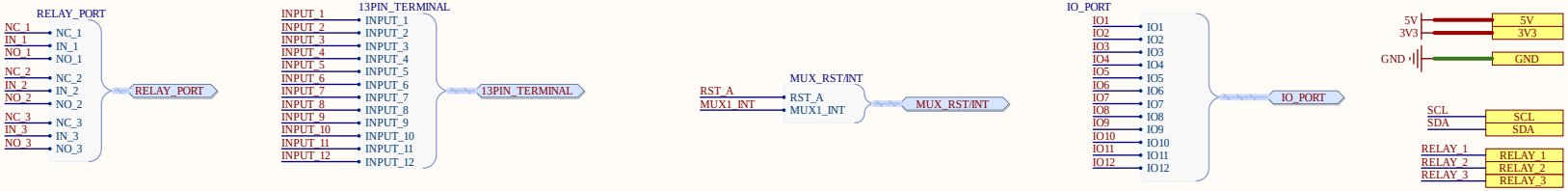


OPTO ISOLATION FOR GPIOs



GPIO PULLUP

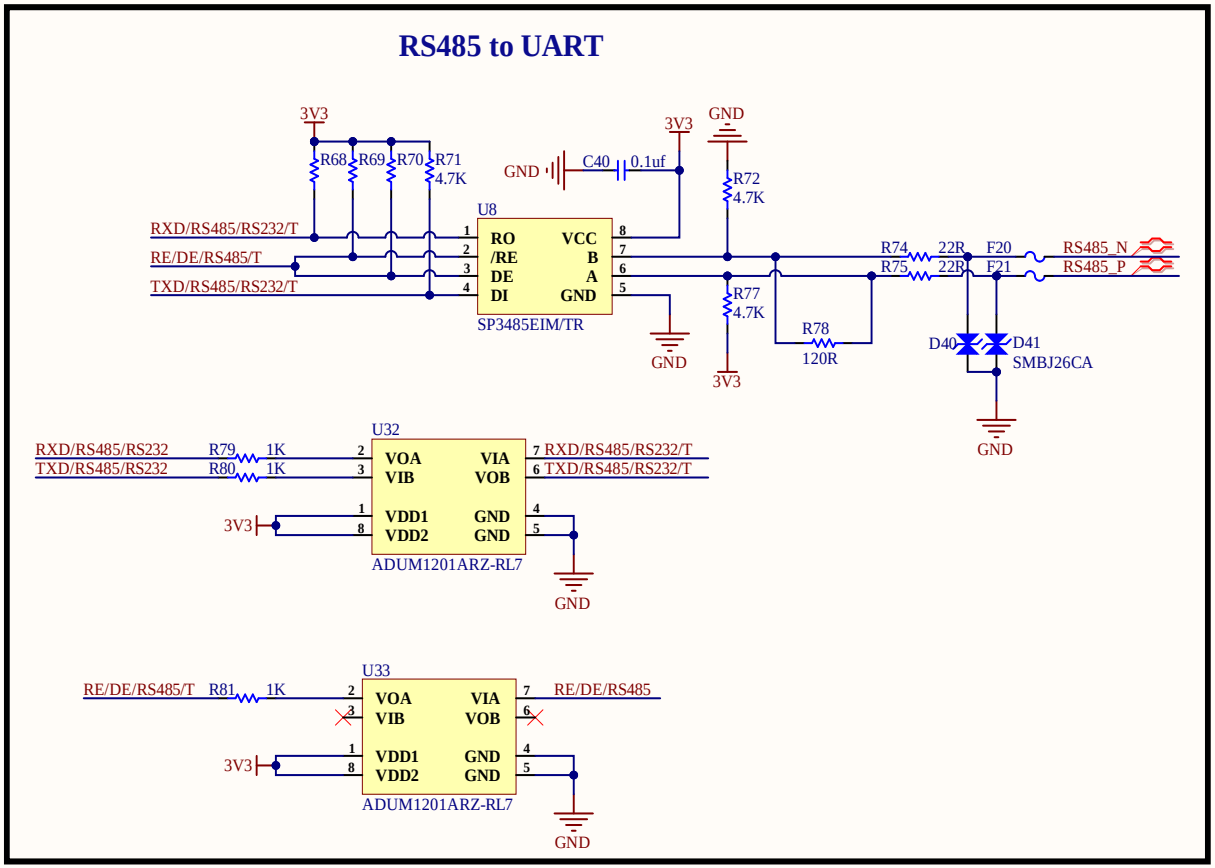




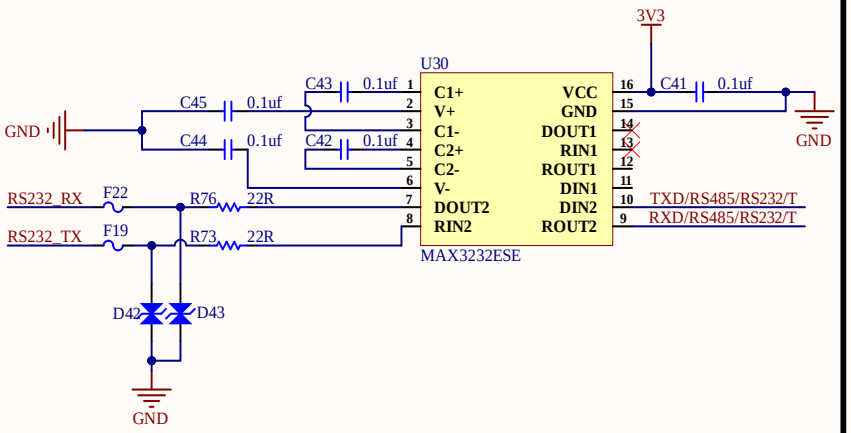
eSpark Consulting Group

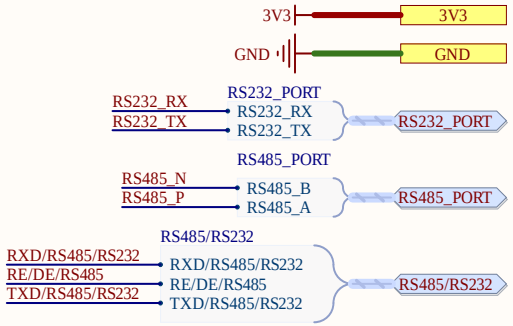
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RS485 to UART



RS232 to UART

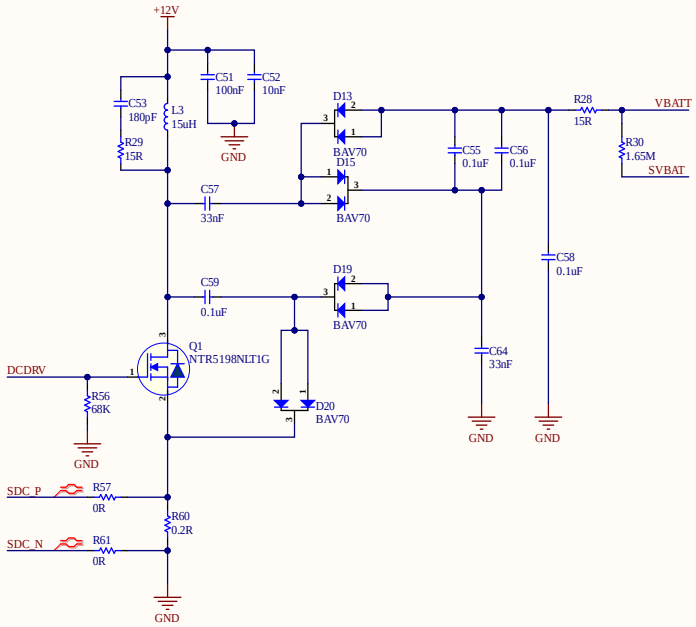




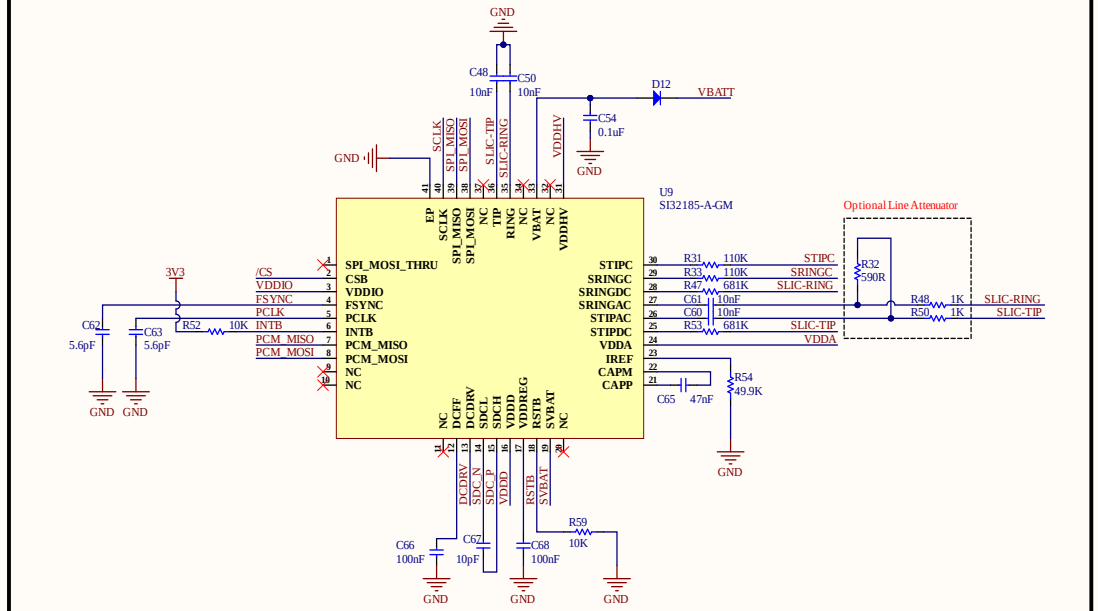
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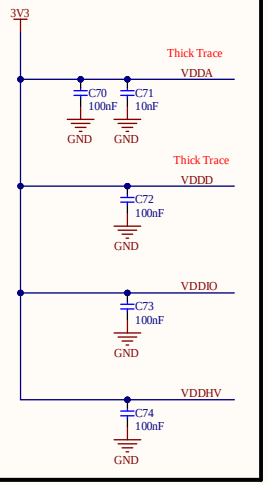
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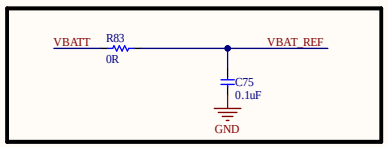


SI32185

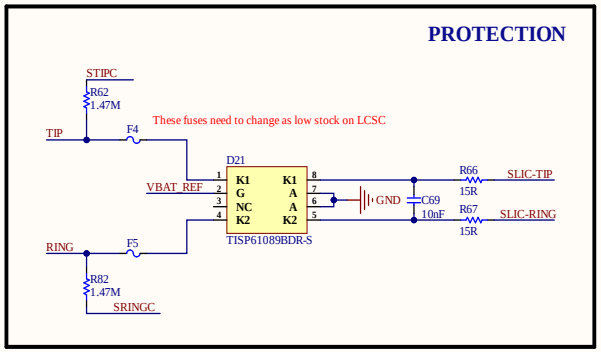


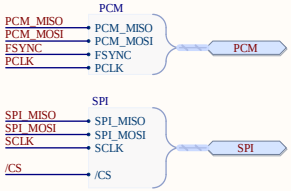
POWER

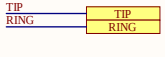


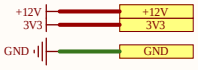


PROTECTION





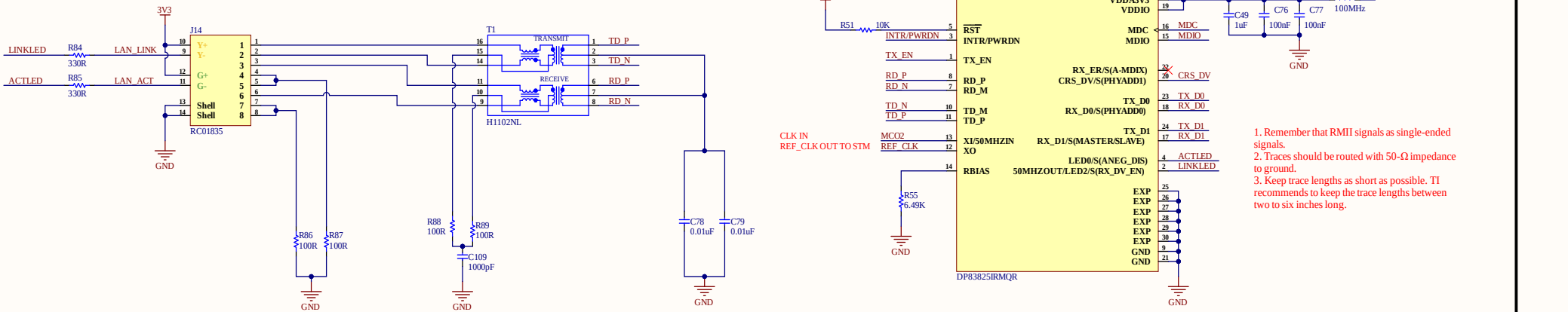


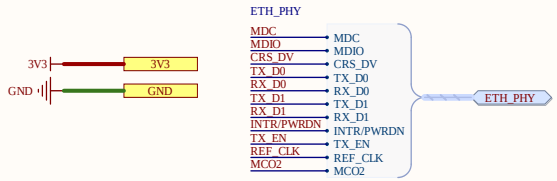


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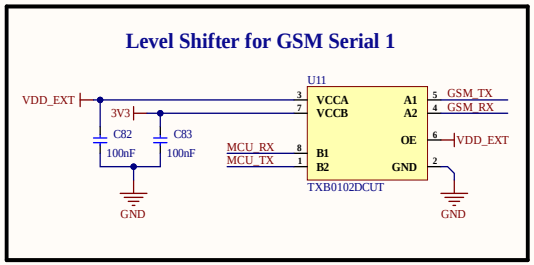
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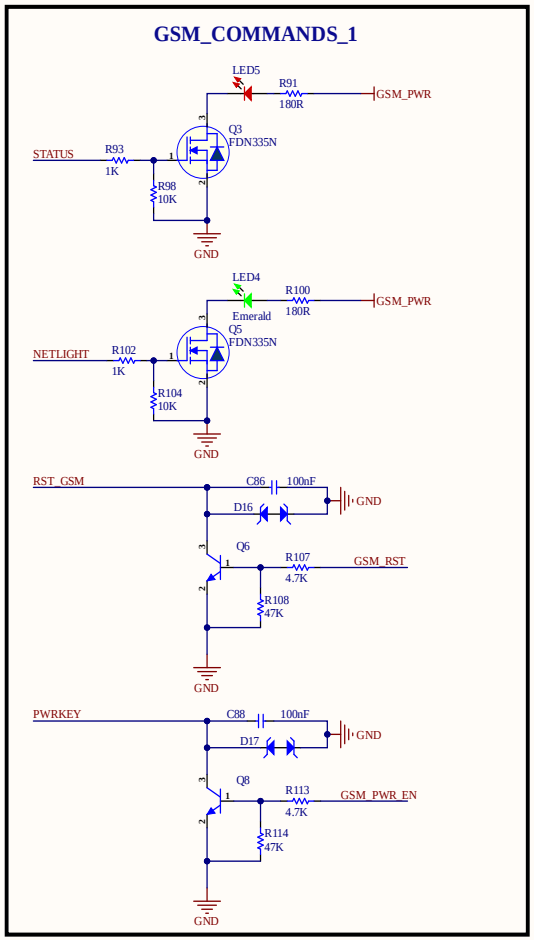
ETHERNET/ DP83825IRMQR PHY

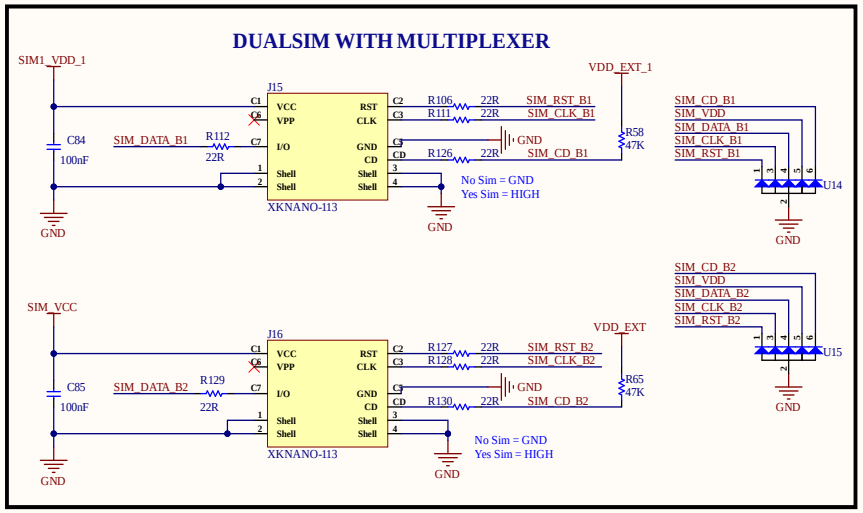


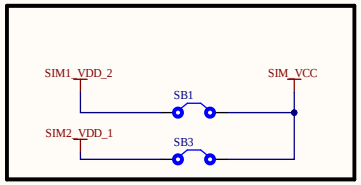


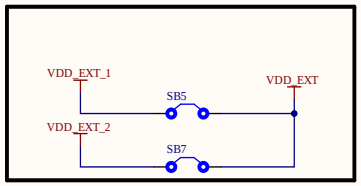
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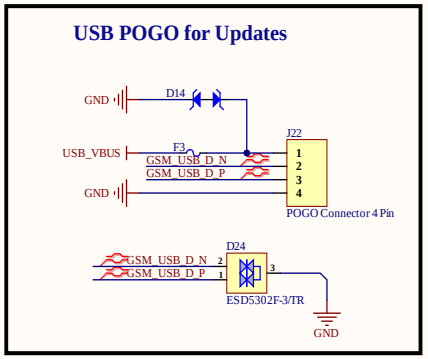




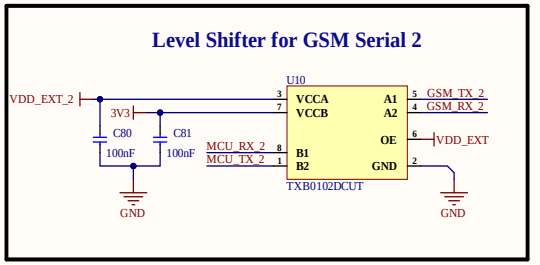


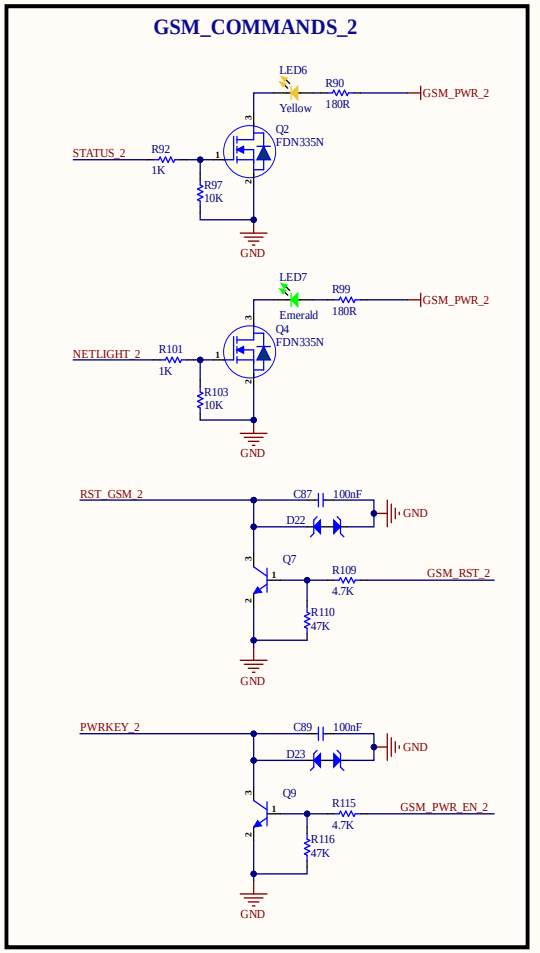


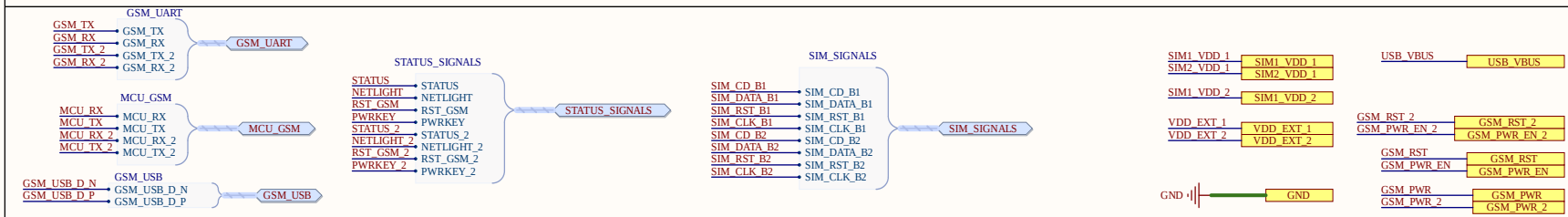




SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommended to follow these guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection
- Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

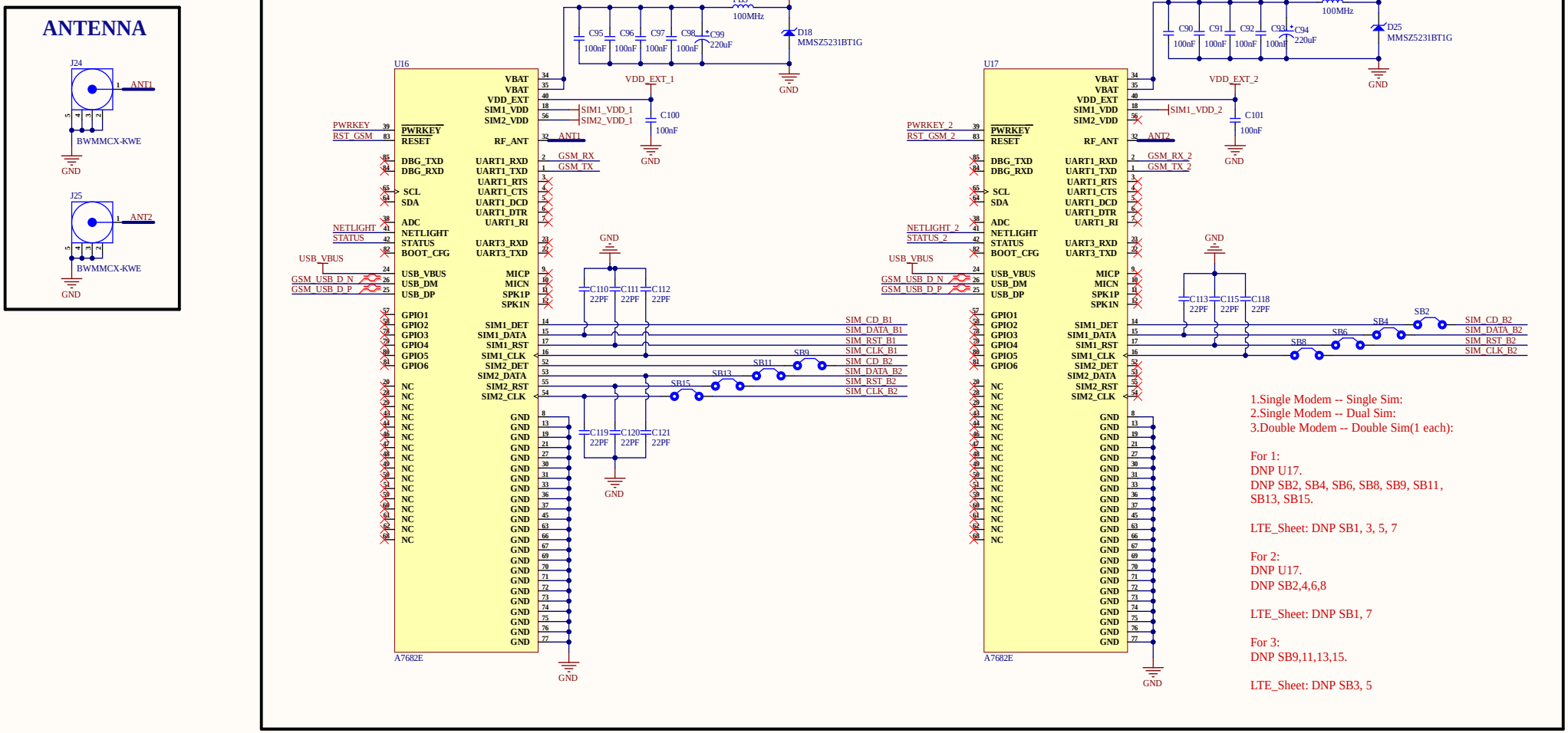




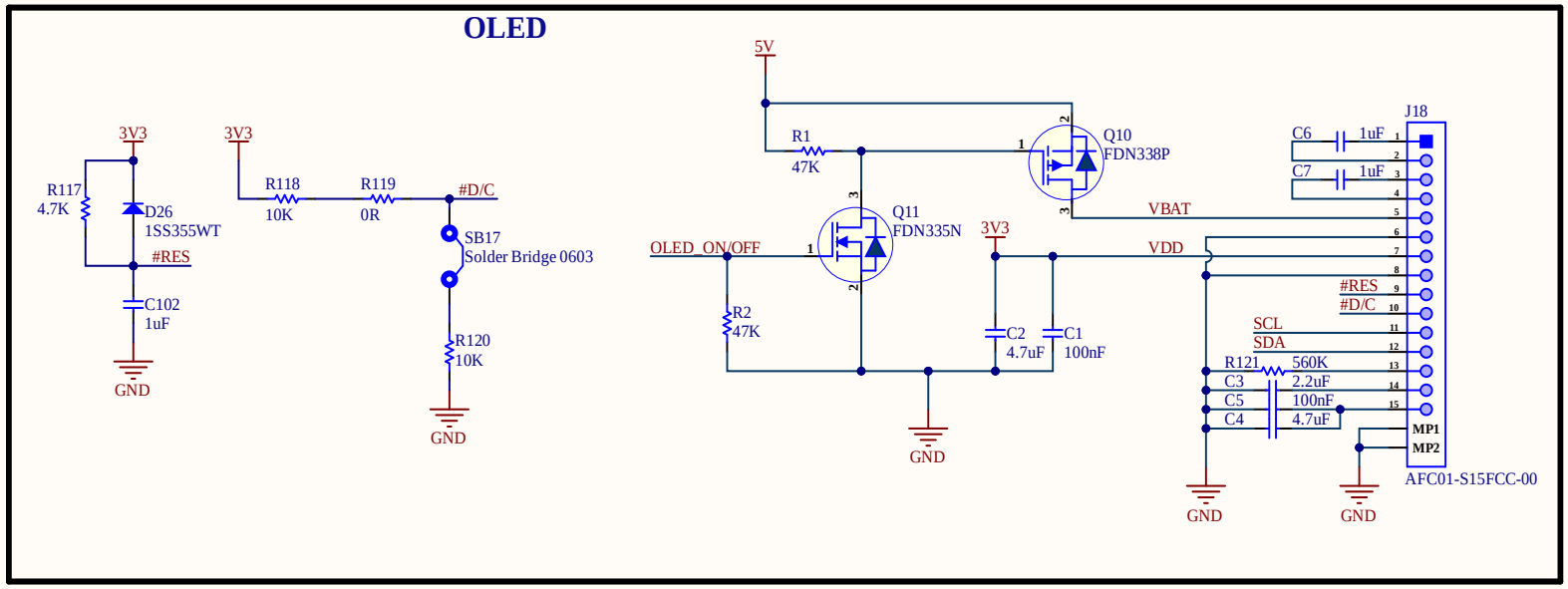


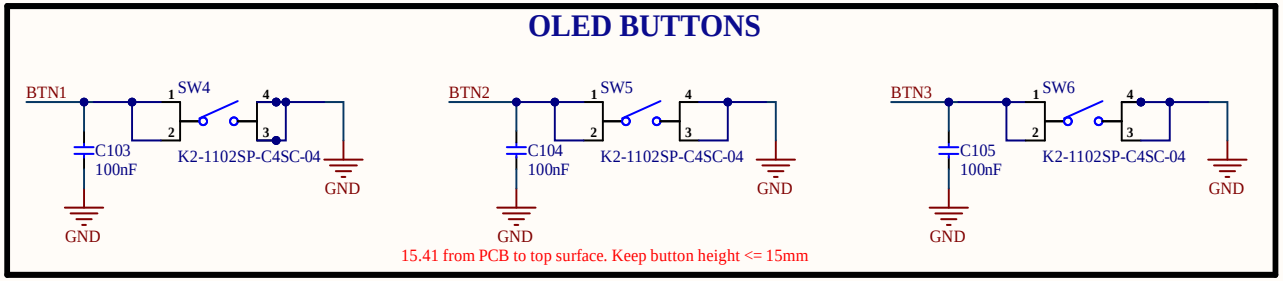
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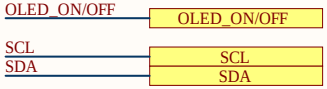
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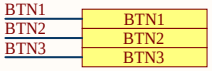


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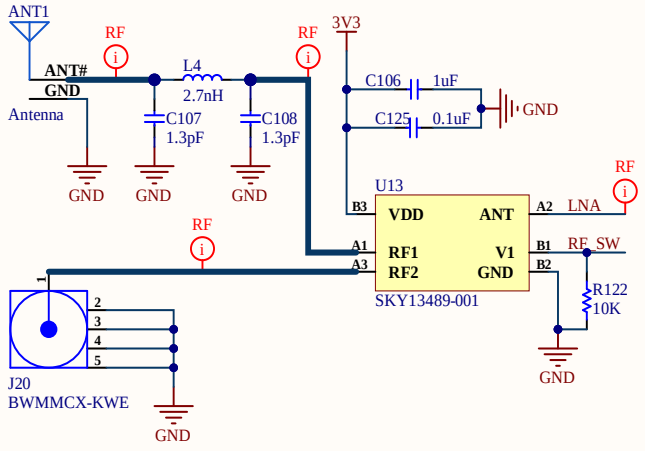


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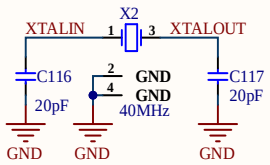
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ESP32-C3FH4

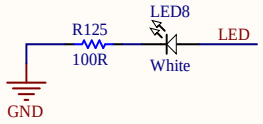
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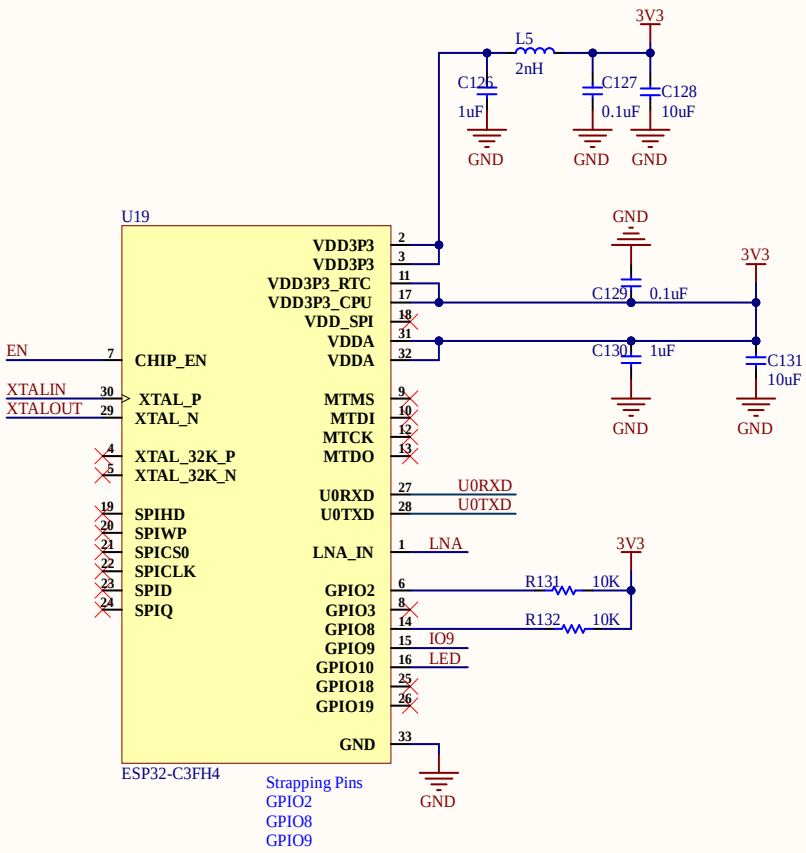


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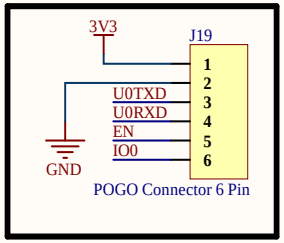


TEST LED

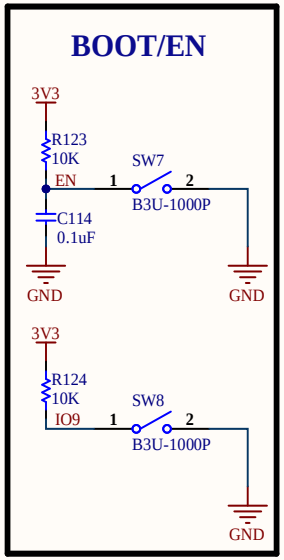


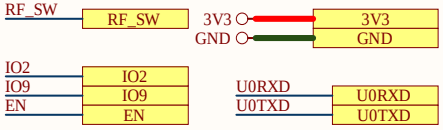


POGO



BOOT/EN





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